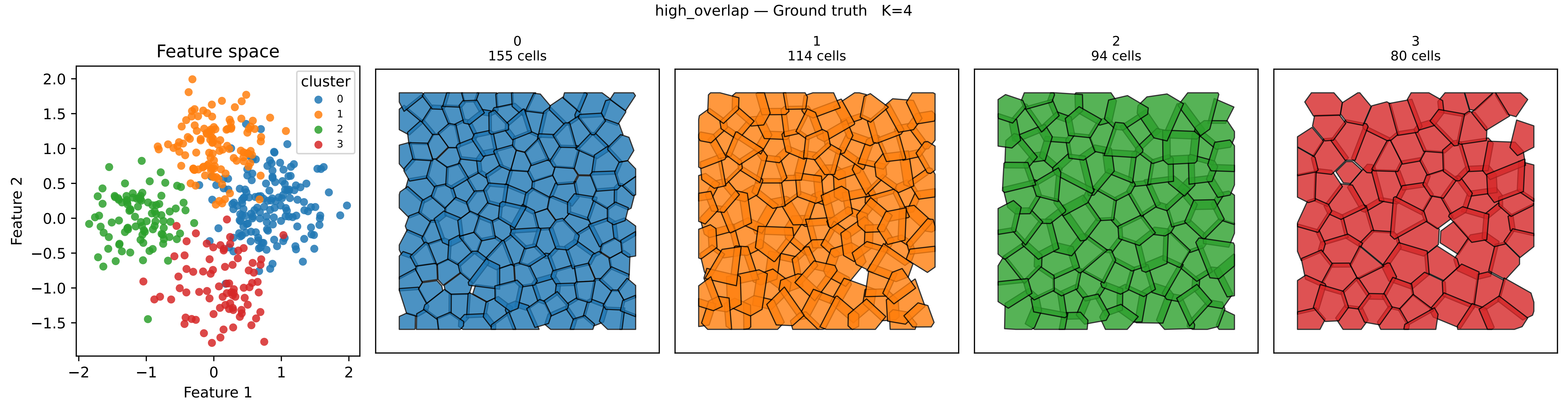
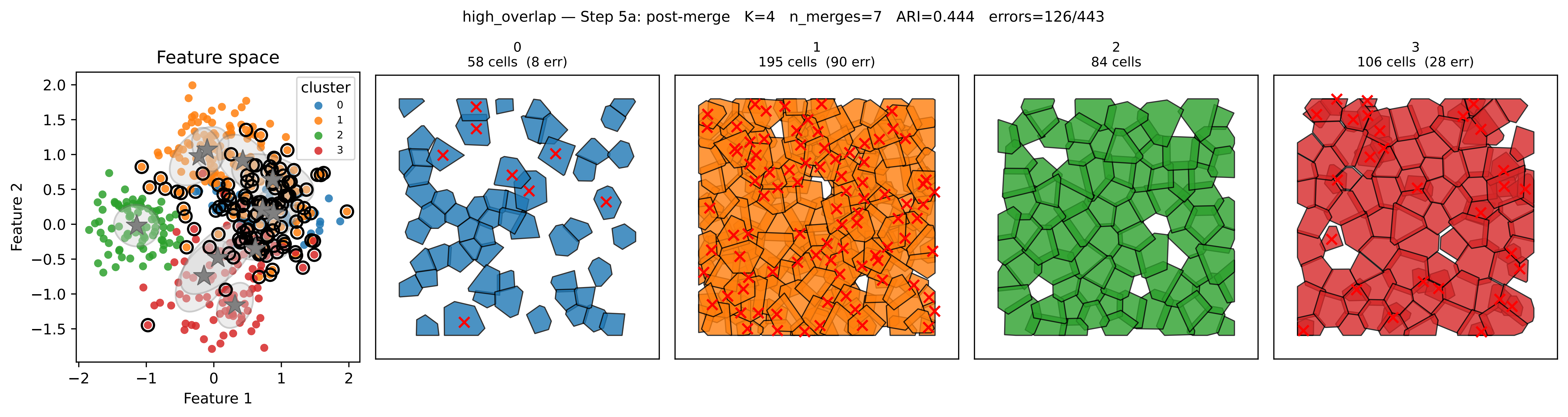


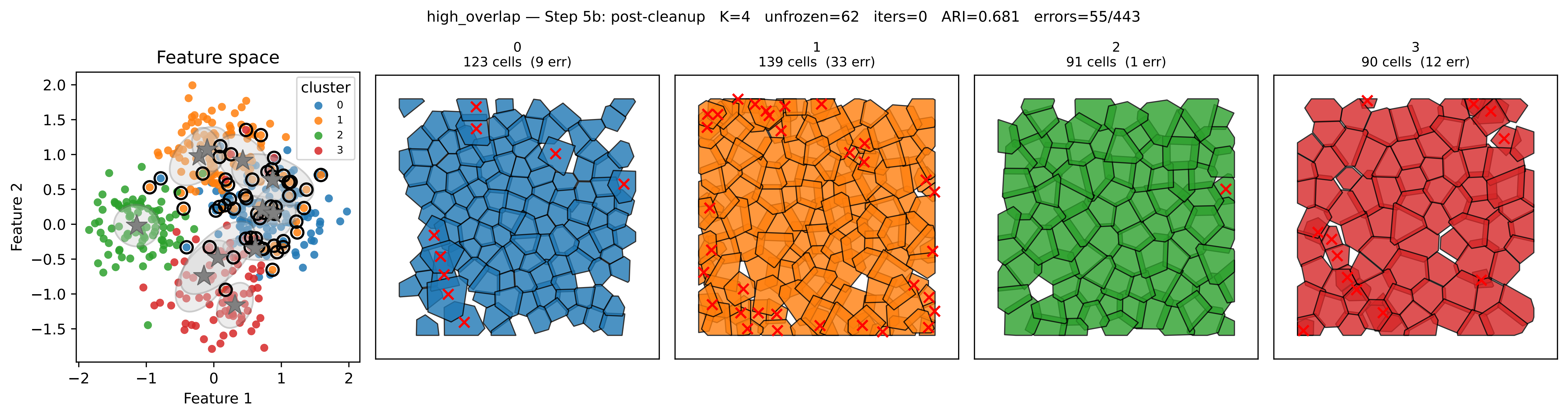
high_overlap — Ground truth K=4
Feature space
cluster
0
1
2
3
Feature 1
Feature 2
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2
−1
0
1
2
0
155 cells
1
114 cells
2
94 cells
3
80 cells

high_overlap — Step 1+2: GMM (init='leiden') K_init=None   ARI=0.373
Feature space
Feature 1
Feature 2
cluster
0
1
2
3
4
5
6
7
8
9
10
0
84 cells
1
43 cells
2
51 cells
3
12 cells
4
60 cells
5
33 cells
6
53 cells
7
26 cells
8
31 cells
9
32 cells
10
18 cells

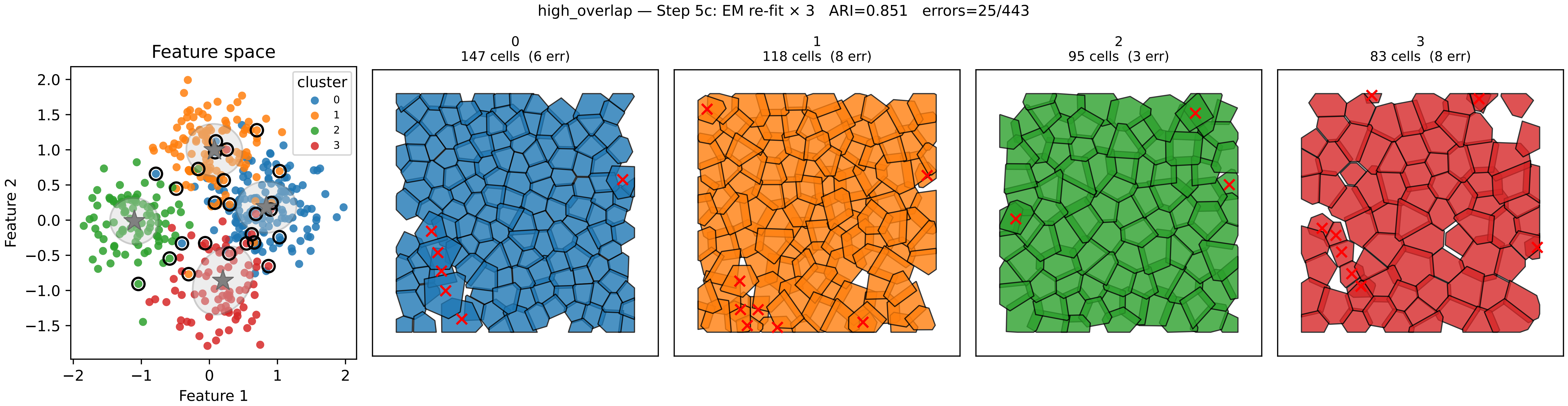
high_overlap — Step 3: post-split K_eff=11 n_splits=0 ARI=0.373

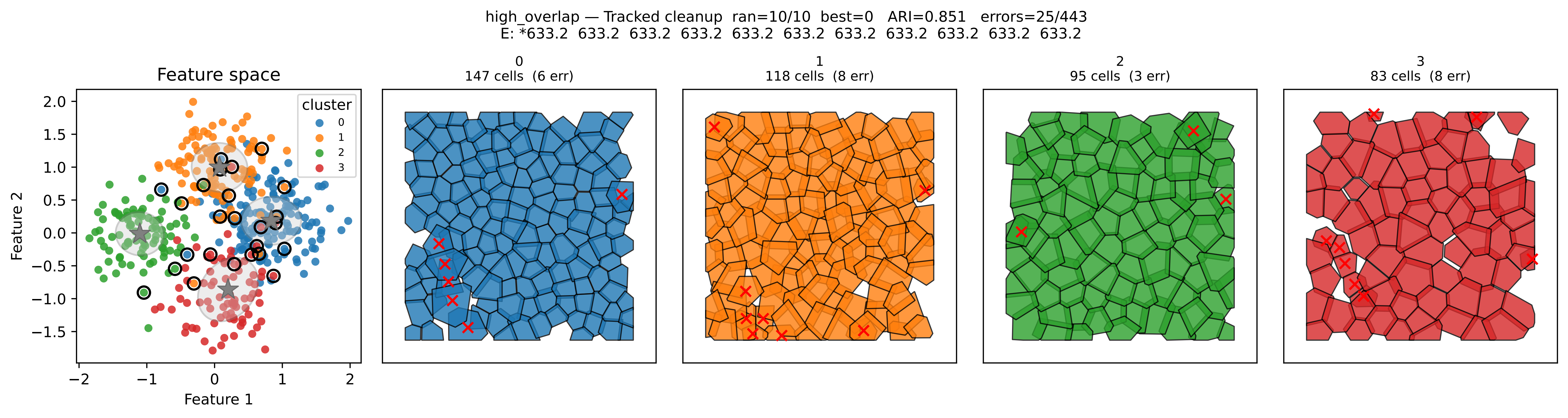
high_overlap — Step 4: post-ICM K_eff=11 iters=4 ARI=0.416
Feature space
cluster
0
1
2
3
4
5
6
7
8
9
10
Feature 1
Feature 2
0
84 cells
1
42 cells
2
48 cells
3
12 cells
4
58 cells
5
58 cells
6
46 cells
7
24 cells
8
28 cells
9
29 cells
10
14 cells

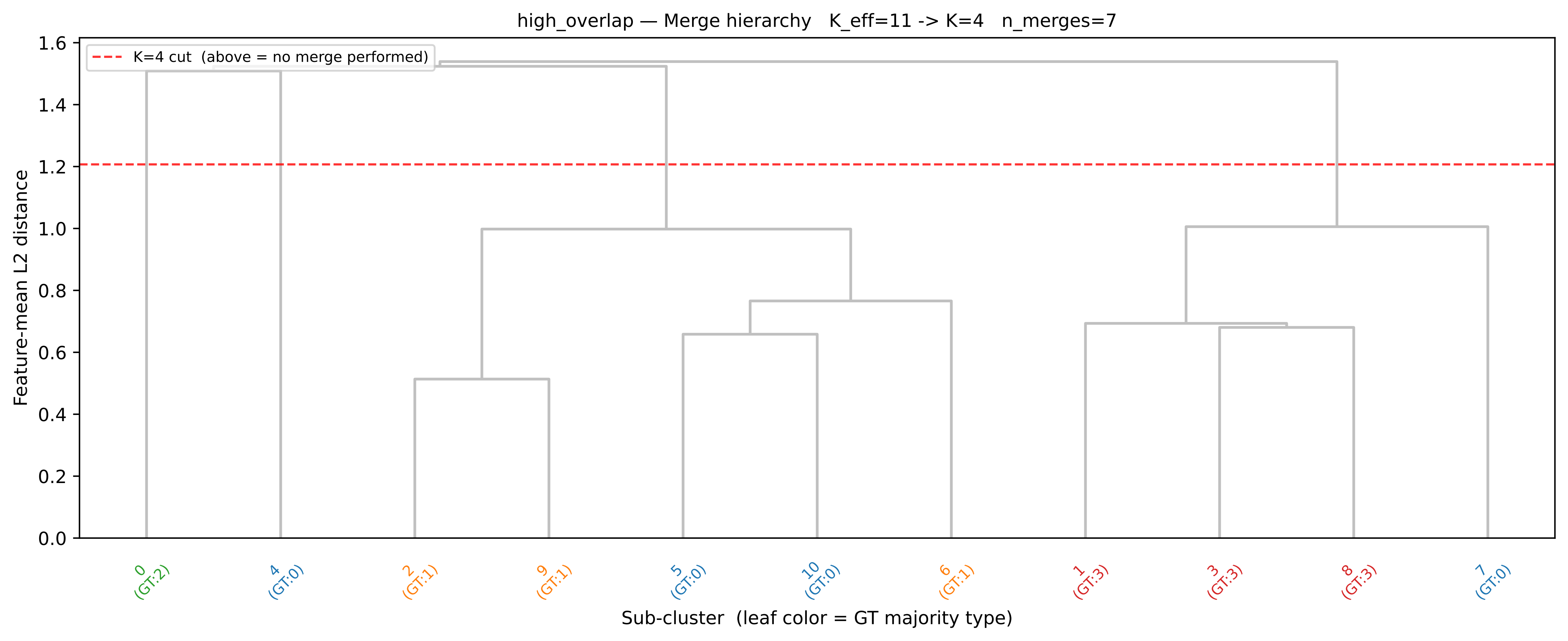

high_overlap — Step 5a: post-merge K=4 n_merges=7 ARI=0.444 errors=126/443
Feature space
cluster
0
1
2
3
Feature 1
Feature 2
0
58 cells (8 err)
1
195 cells (90 err)
2
84 cells
3
106 cells (28 err)



high_overlap — Step 5b: post-cleanup K=4 unfrozen=62 iters=0 ARI=0.681 errors=55/443
Feature space
cluster
0
1
2
3
Feature 1
Feature 2
0
123 cells (9 err)
1
139 cells (33 err)
2
91 cells (1 err)
3
90 cells (12 err)


high_overlap — Step 5c: EM re-fit × 3 ARI=0.851 errors=25/443
Feature space
cluster
0
1
2
3
Feature 1
Feature 2
0
147 cells (6 err)
1
118 cells (8 err)
2
95 cells (3 err)
3
83 cells (8 err)


high_overlap — Tracked cleanup ran=10/10 best=0 ARI=0.851 errors=25/443
E: *633.2 633.2 633.2 633.2 633.2 633.2 633.2 633.2 633.2 633.2 633.2
Feature space
cluster
0
1
2
3
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2
−1
0
1
2
Feature 1
Feature 2
0
147 cells (6 err)
1
118 cells (8 err)
2
95 cells (3 err)
3
83 cells (8 err)


high_overlap — Merge hierarchy K_eff=11 -> K=4 n_merges=7
K=4 cut (above = no merge performed)
Feature-mean L2 distance
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0 (GT:2)
4 (GT:0)
2 (GT:1)
9 (GT:1)
5 (GT:0)
10 (GT:0)
6 (GT:1)
1 (GT:3)
3 (GT:3)
8 (GT:3)
7 (GT:0)
Sub-cluster (leaf color = GT majority type)